

本次作業是使用 System C 實作 AlexNet，並將 HW1 的模型進一步模組化，分別使用 sc-signal、sc-buffer、sc-fifo 三種 channel 來完成模組之間的資料傳遞，其推論結果要和 HW1 一致並能正確地輸出 top 100 類別和它的機率。

模型架構和資料流程：

一樣使用的模型是 AlexNet，同 HW1，在這次實作將整體系統主要分成 2 個部分：

- 1. Pattern 模組：負責讀取輸入影像，送出控制訊號，接收輸出結果
 - 2. AlexNet 模組：負責執行完整的 AlexNet forward inference
- 在 sc-signal 和 sc-buffer 中，Pattern 和 AlexNet 之間直接透過對應的 channel 連接；在 sc-fifo 中，因為 Pattern 本身是 signal-based 介面，因此額外設計 bridge module，將 signal 和 fifo 之間做轉換，讓整體系統仍能使用 sc-fifo 進行資料傳輸。

Pattern $\rightarrow$ Channel $\rightarrow$ AlexNet $\rightarrow$ Channel $\rightarrow$ Pattern

實作方法：

1. 輸入資料處理：輸入影像 cat.txt 和 dog.txt 讀入，讀入方式將一維輸入資料還原成 $3 \times 224 \times 224$ 的影像 tensor，接著進行 zero padding 補成 $3 \times 227 \times 227$
2. 權重和偏差載入：AlexNet 每一層的權重和 bias 都由助教提供的 txt 檔讀入。程式在第一次執行 inference 時會先將所有層的參數載入記憶體，之後再依序執行 forward computation。
3. Forward inference 流程：
Conv 1 $\rightarrow$ ReLU $\rightarrow$ Pool 1
Conv 2 $\rightarrow$ ReLU $\rightarrow$ Pool 2
Conv 3 $\rightarrow$ ReLU
Conv 4 $\rightarrow$ ReLU
Conv 5 $\rightarrow$ ReLU $\rightarrow$ Pool 5

Flatten
FC6 → ReLU
FC7 → ReLU
FC8
Softmax

4. 三種版本的製作方式：

(1) `sc-signal`：用 `sc-signal` 作為 Pattern 和 AlexNet 之間的主要通訊方式。rst、in_valid、out_valid 等控制訊號以及 image/output 資料向量皆透過 `sc-signal` 連接。

(2) `sc-buffer`：整體架構和 `sc-signal` 相近，但中間通道改成 `sc-buffer`。由於 `sc-buffer` 即使寫入相同值也會更新，因此行為上還是和 `sc-signal` 有不同，但對這次作業結果不影響。

(3) `sc-fifo`：使用 `sc-fifo` 作為主要資料通道。由於原始 Pattern 模組介面仍然是 signal 型式，因此額外加入 bridge modules，負責：

- 將 Pattern 輸出的 image 和 valid signal 轉成 fifo 輸入
- 將 AlexNet 的輸出結果轉成 fifo，再送回 Pattern

Simulation Results：

1. `sc-signal`
(cat)

```
2:48 es26m1chip70@ee241~/.hw2/01_SOURCE/SC_SIGNAL$ make clean
rm -rf run
2:48 es26m1chip70@ee241~/.hw2/01_SOURCE/SC_SIGNAL$ make
g++ -f -I /RAID2/COURSE/2026_Spring/es26m1chip/es26m1chipT10/systemc-2.3.3/include -I -I /RAID2/COURSE/2026_Spring/es26m1chip/es26m1chipT10/systemc-2.3.3/lib/linux64 -g run *.cpp -lsystemc-2.3.3 -Wl,-rpath,/RAID2/COURSE/2026_Spring/es26m1chip/es26m1chipT10/systemc-2.3.3/lib/linux64 -g
2:48 es26m1chip70@ee241~/.hw2/01_SOURCE/SC_SIGNAL$ ./run cat.txt

SystemC 2.3.3-Accellera --- Mar 3 2026 15:59:48
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED

Top 100 classes:
=====
idx | val | possibility | class name
-----
285 | 20.21 | 06.28 | Egyptian cat
281 | 16.14 | 1.65 | tabby
282 | 15.73 | 1.10 | tiger cat
287 | 14.79 | 0.43 | lynx
728 | 14.41 | 0.29 | plastic bag
330 | 12.72 | 0.05 | wood rabbit
331 | 12.19 | 0.03 | hare
457 | 10.84 | 0.01 | bow tie
335 | 10.67 | 0.01 | fox squirrel
463 | 10.57 | 0.01 | bucket
436 | 10.32 | 0.00 | carton
876 | 10.29 | 0.00 | tub
602 | 10.18 | 0.00 | lens cap
904 | 10.01 | 0.00 | window screen
790 | 9.56 | 0.00 | paper towel
278 | 9.38 | 0.00 | kit fox
8 | 9.29 | 0.00 | hen
204 | 9.21 | 0.00 | Siamese cat
722 | 8.80 | 0.00 | ping-pong ball
434 | 8.77 | 0.00 | bath towel
452 | 8.71 | 0.00 | bonnet
289 | 8.31 | 0.00 | snow leopard
753 | 8.30 | 0.00 | radiator
681 | 8.11 | 0.00 | notebook
673 | 8.04 | 0.00 | mouse
840 | 8.03 | 0.00 | swab
773 | 7.94 | 0.00 | saltshaker
782 | 7.90 | 0.00 | screen
896 | 7.71 | 0.00 | washbasin
44 | 7.70 | 0.00 | alligator lizard
719 | 7.69 | 0.00 | piggy bank
138 | 7.68 | 0.00 | bustard
908 | 7.57 | 0.00 | rad
543 | 7.52 | 0.00 | dumbbell
451 | 7.40 | 0.00 | bolo tie
283 | 7.40 | 0.00 | Persian cat
272 | 7.32 | 0.00 | coyote
889 | 7.27 | 0.00 | toilet tissue
769 | 7.23 | 0.00 | rule
713 | 7.14 | 0.00 | photocopier
```

(dog)

```
2:48 es26m1chip70@ee241~/.hw2/01_SOURCE/SC_SIGNAL$ ./run dog.txt

SystemC 2.3.3-Accellera --- Mar 3 2026 15:59:48
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED

Top 100 classes:
=====
idx | val | possibility | class name
-----
207 | 16.59 | 38.63 | golden retriever
175 | 15.57 | 13.86 | otterhound
220 | 15.36 | 11.26 | Sussex spaniel
163 | 15.00 | 7.86 | bloodhound
219 | 14.59 | 5.22 | cocker spaniel
168 | 14.39 | 4.28 | redbone
160 | 14.35 | 4.07 | Afghan hound
213 | 14.18 | 3.46 | Irish setter
291 | 14.10 | 3.19 | lion
211 | 13.01 | 1.07 | vizsla
244 | 12.81 | 0.88 | Tibetan mastiff
216 | 12.69 | 0.78 | lumber
200 | 12.46 | 0.62 | Tibetan terrier
159 | 12.42 | 0.59 | Rhodesian ridgeback
152 | 12.38 | 0.57 | Japanese spaniel
167 | 12.01 | 0.39 | English foxhound
208 | 11.65 | 0.28 | Labrador retriever
294 | 11.63 | 0.27 | brown bear
165 | 11.51 | 0.24 | black-and-tan coonhound
262 | 11.46 | 0.23 | Brabancon griffon
156 | 11.40 | 0.22 | Blenheim spaniel
185 | 11.29 | 0.19 | Norfolk terrier
260 | 11.11 | 0.16 | chow
267 | 11.08 | 0.16 | standard poodle
395 | 10.87 | 0.13 | orangutan
154 | 10.74 | 0.11 | Pekinese
368 | 10.70 | 0.11 | gibbon
215 | 10.65 | 0.10 | Brittany spaniel
161 | 10.59 | 0.10 | basset
214 | 10.47 | 0.08 | Gordon setter
176 | 10.39 | 0.08 | Saluki
212 | 10.31 | 0.07 | English setter
852 | 9.88 | 0.05 | tennis ball
255 | 9.87 | 0.05 | Leonberg
194 | 9.77 | 0.04 | Dandie Dinmont
672 | 9.59 | 0.04 | mountain tent
274 | 9.44 | 0.03 | dhole
335 | 9.44 | 0.03 | fox squirrel
371 | 9.40 | 0.03 | patas
190 | 9.31 | 0.03 | Sealyham terrier
384 | 9.29 | 0.03 | indri
193 | 9.15 | 0.02 | Australian terrier
166 | 9.00 | 0.02 | Walker hound
206 | 8.92 | 0.02 | curly-coated retriever
380 | 8.90 | 0.02 | titi
170 | 8.86 | 0.02 | Irish wolfhound
191 | 8.85 | 0.02 | Airedale
```

2. sc-buffer

(cat)

```
2:53 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_FIFO]$ cd ~/hw2/01_SOURCE/SC_BUFFER
2:56 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_BUFFER]$ nano alexnet_buffer.cpp
2:59 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_BUFFER]$ make clean
rm -rf run
2:59 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_BUFFER]$ make
g++ -I . -I /RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/include -L . -L /RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/lib-linux64 -o run *.cpp -lsystemc-2.3.3 -Wl,-rpath,/RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/lib-linux64 -g
2:59 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_BUFFER]$ ./run cat.txt

SystemC 2.3.3-Accellera --- Mar 3 2026 15:59:48
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED

Top 100 classes:
=====
idx | val | possibility | class name
-----
285 | 20.21 | 96.38 | Egyptian cat
281 | 16.14 | 1.65 | tabby
282 | 15.73 | 1.10 | tiger cat
287 | 14.79 | 0.43 | lynx
728 | 14.41 | 0.29 | plastic bag
330 | 12.73 | 0.05 | wood rabbit
331 | 12.19 | 0.03 | hare
457 | 10.94 | 0.01 | bow tie
335 | 10.67 | 0.01 | fox squirrel
463 | 10.57 | 0.01 | bucket
478 | 10.32 | 0.00 | carton
876 | 10.29 | 0.00 | tub
622 | 10.18 | 0.00 | lens cap
904 | 10.01 | 0.00 | window screen
700 | 9.56 | 0.00 | paper towel
278 | 9.39 | 0.00 | kit fox
8 | 9.29 | 0.00 | hen
284 | 9.21 | 0.00 | Siamese cat
722 | 8.80 | 0.00 | ping-pong ball
434 | 8.77 | 0.00 | bath towel
452 | 8.71 | 0.00 | bonnet
289 | 8.31 | 0.00 | snow leopard
753 | 8.30 | 0.00 | radiator
681 | 8.11 | 0.00 | notebook
673 | 8.04 | 0.00 | mouse
840 | 8.03 | 0.00 | swab
773 | 7.94 | 0.00 | saltshaker
782 | 7.90 | 0.00 | screen
896 | 7.71 | 0.00 | washbasin
44 | 7.70 | 0.00 | alligator lizard
719 | 7.69 | 0.00 | piggy bank
138 | 7.68 | 0.00 | bustard
998 | 7.57 | 0.00 | ear
543 | 7.52 | 0.00 | dumbbell
451 | 7.49 | 0.00 | bolo tie
283 | 7.40 | 0.00 | Persian cat
272 | 7.32 | 0.00 | coyote
999 | 7.27 | 0.00 | toilet tissue
769 | 7.23 | 0.00 | rule
713 | 7.14 | 0.00 | photocopier
```

(dog)

```
2:59 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_BUFFER]$ ./run dog.txt

SystemC 2.3.3-Accellera --- Mar 3 2026 15:59:48
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED

Top 100 classes:
=====
idx | val | possibility | class name
-----
207 | 16.59 | 38.63 | golden retriever
175 | 15.57 | 13.86 | otterhound
220 | 15.36 | 11.26 | Sussex spaniel
163 | 15.00 | 7.86 | bloodhound
219 | 14.59 | 5.22 | cocker spaniel
168 | 14.39 | 4.28 | redbone
160 | 14.35 | 4.07 | Afghan hound
213 | 14.18 | 3.46 | Irish setter
291 | 14.10 | 3.19 | lion
211 | 13.01 | 1.07 | vizsla
244 | 12.81 | 0.88 | Tibetan mastiff
216 | 12.69 | 0.78 | clumber
200 | 12.46 | 0.62 | Tibetan terrier
159 | 12.42 | 0.59 | Rhodesian ridgeback
152 | 12.38 | 0.57 | Japanese spaniel
167 | 12.01 | 0.39 | English foxhound
208 | 11.65 | 0.28 | Labrador retriever
294 | 11.63 | 0.27 | brown bear
165 | 11.51 | 0.24 | black-and-tan coonhound
262 | 11.46 | 0.23 | Brabancon griffon
156 | 11.40 | 0.21 | Blenheim spaniel
185 | 11.29 | 0.19 | Norfolk terrier
260 | 11.11 | 0.16 | chow
267 | 11.08 | 0.16 | standard poodle
365 | 10.87 | 0.13 | orangutan
154 | 10.74 | 0.11 | Pekinese
368 | 10.70 | 0.11 | gibbon
215 | 10.66 | 0.10 | Brittany spaniel
161 | 10.59 | 0.10 | basset
214 | 10.47 | 0.08 | Gordon setter
176 | 10.39 | 0.08 | Saluki
212 | 10.31 | 0.07 | English setter
852 | 9.88 | 0.05 | tennis ball
255 | 9.87 | 0.05 | Leonberg
194 | 9.77 | 0.04 | Dandie Dinmont
672 | 9.59 | 0.04 | mountain tent
274 | 9.44 | 0.03 | dhole
335 | 9.44 | 0.03 | fox squirrel
371 | 9.40 | 0.03 | patas
190 | 9.31 | 0.03 | Sealyham terrier
384 | 9.29 | 0.03 | indri
193 | 9.15 | 0.02 | Australian terrier
166 | 9.00 | 0.02 | Walker hound
206 | 8.92 | 0.02 | curly-coated retriever
380 | 8.90 | 0.02 | titi
170 | 8.86 | 0.02 | Irish wolfhound
191 | 8.85 | 0.02 | Airedale
```

3. sc-fifo

(cat)

```
3:02 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_BUFFER]$ cd ~/hw2/01_SOURCE/SC_FIFO
3:02 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_FIFO]$ nano alexnet_fifo.cpp
3:12 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_FIFO]$ make clean
rm -rf run
3:12 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_FIFO]$ make
g++ -I . -I /RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/include -L . -L /RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/lib-linux64 -o run *.cpp -lsystemc-2.3.3 -Wl,-rpath,/RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/lib-linux64 -g
3:12 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_FIFO]$ ./run cat.txt

SystemC 2.3.3-Accellera --- Mar 3 2026 15:59:48
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED

Top 100 classes:
=====
idx | val | possibility | class name
-----
285 | 20.21 | 96.38 | Egyptian cat
281 | 16.14 | 1.65 | tabby
282 | 15.73 | 1.10 | tiger cat
287 | 14.79 | 0.43 | lynx
728 | 14.41 | 0.29 | plastic bag
330 | 12.73 | 0.05 | wood rabbit
331 | 12.19 | 0.03 | hare
457 | 10.94 | 0.01 | bow tie
335 | 10.67 | 0.01 | fox squirrel
463 | 10.57 | 0.01 | bucket
478 | 10.32 | 0.00 | carton
876 | 10.29 | 0.00 | tub
622 | 10.18 | 0.00 | lens cap
904 | 10.01 | 0.00 | window screen
700 | 9.56 | 0.00 | paper towel
278 | 9.39 | 0.00 | kit fox
8 | 9.29 | 0.00 | hen
284 | 9.21 | 0.00 | Siamese cat
722 | 8.80 | 0.00 | ping-pong ball
434 | 8.77 | 0.00 | bath towel
452 | 8.71 | 0.00 | bonnet
289 | 8.31 | 0.00 | snow leopard
753 | 8.30 | 0.00 | radiator
681 | 8.11 | 0.00 | notebook
673 | 8.04 | 0.00 | mouse
840 | 8.03 | 0.00 | swab
773 | 7.94 | 0.00 | saltshaker
782 | 7.90 | 0.00 | screen
896 | 7.71 | 0.00 | washbasin
44 | 7.70 | 0.00 | alligator lizard
719 | 7.69 | 0.00 | piggy bank
138 | 7.68 | 0.00 | bustard
998 | 7.57 | 0.00 | ear
543 | 7.52 | 0.00 | dumbbell
451 | 7.49 | 0.00 | bolo tie
283 | 7.40 | 0.00 | Persian cat
272 | 7.32 | 0.00 | coyote
999 | 7.27 | 0.00 | toilet tissue
```

(dog)

```
3:13 es26mlchip070@ee24[~/hw2/01_SOURCE/SC_FIFO]$ ./run dog.txt

SystemC 2.3.3-Accellera --- Mar 3 2026 15:59:48
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED

Top 100 classes:
=====
idx | val | possibility | class name
-----
207 | 16.59 | 38.63 | golden retriever
175 | 15.57 | 13.86 | otterhound
220 | 15.36 | 11.26 | Sussex spaniel
163 | 15.00 | 7.86 | bloodhound
219 | 14.59 | 5.22 | cocker spaniel
168 | 14.39 | 4.28 | redbone
160 | 14.35 | 4.07 | Afghan hound
213 | 14.18 | 3.46 | Irish setter
291 | 14.10 | 3.19 | lion
211 | 13.01 | 1.07 | vizsla
244 | 12.81 | 0.88 | Tibetan mastiff
216 | 12.69 | 0.78 | clumber
200 | 12.46 | 0.62 | Tibetan terrier
159 | 12.42 | 0.59 | Rhodesian ridgeback
152 | 12.38 | 0.57 | Japanese spaniel
167 | 12.01 | 0.39 | English foxhound
208 | 11.65 | 0.28 | Labrador retriever
294 | 11.63 | 0.27 | brown bear
165 | 11.51 | 0.24 | black-and-tan coonhound
262 | 11.46 | 0.23 | Brabancon griffon
156 | 11.40 | 0.21 | Blenheim spaniel
185 | 11.29 | 0.19 | Norfolk terrier
260 | 11.11 | 0.16 | chow
267 | 11.08 | 0.16 | standard poodle
365 | 10.87 | 0.13 | orangutan
154 | 10.74 | 0.11 | Pekinese
368 | 10.70 | 0.11 | gibbon
215 | 10.66 | 0.10 | Brittany spaniel
161 | 10.59 | 0.10 | basset
214 | 10.47 | 0.08 | Gordon setter
176 | 10.39 | 0.08 | Saluki
212 | 10.31 | 0.07 | English setter
852 | 9.88 | 0.05 | tennis ball
255 | 9.87 | 0.05 | Leonberg
194 | 9.77 | 0.04 | Dandie Dinmont
672 | 9.59 | 0.04 | mountain tent
274 | 9.44 | 0.03 | dhole
335 | 9.44 | 0.03 | fox squirrel
371 | 9.40 | 0.03 | patas
190 | 9.31 | 0.03 | Sealyham terrier
384 | 9.29 | 0.03 | indri
193 | 9.15 | 0.02 | Australian terrier
166 | 9.00 | 0.02 | Walker hound
206 | 8.92 | 0.02 | curly-coated retriever
380 | 8.90 | 0.02 | titi
170 | 8.86 | 0.02 | Irish wolfhound
191 | 8.85 | 0.02 | Airedale
379 | 8.84 | 0.02 | howler monkey
```

Challenges :

1. 檔案路徑錯誤，一開始在 AlexNet.cpp 中讀取 weight 和 bias 時，使用了錯誤的路徑，導致程式雖然能執行，但實際上沒有正確讀到模型參數，最後輸出變成平均分配的 softmax 結果，之後修成正確的相對路徑模型輸出才正常。
2. 記憶體使用量過大，由於 AlexNet 模組中包含大量陣列，尤其是全連接層的權重矩陣非常大，如果直接將模組建立成區域變數，造成 segmentation fault，後來改成動態模組 (new) 建立模組，問題就解決了。
3. FIFO 比另外 2 個版本 signal 和 buffer 複雜，因為 Pattern 是 signal-based 介面，不能直接和 sc_fifo 相連，因此需要額外設計 bridge modules 來完成資料轉換。

三種 Channel 差異分析：

1. sc-signal：最基本、直接的通訊方式。

優：架構簡單，好理解，容易和 testbench 整合

缺：沒有特別適合串流式的資料傳輸，資料流程複雜時
彈性小

2. sc-buffer：和 sc-signal 非常相似，但差別在於即使寫入相同值，資料也會更新

優：和 sc-signal 相近，需重複寫入相同值時仍會觸發更新

缺：整體使用方向仍偏向 signal-style communication，
在這次作業中和 sc-signal 差別不大

3. sc_fifo：是佇列型 channel，比較適合 producer-consumer 或 stream-based communication

優：適合資料流、串流式架構，生產者和消費者關係清楚，
更能展現 channel-based communication 的概念

缺：設計複雜，需更多同步和橋接處理，和原本的 signal-based testbench 整合較麻煩

